

Sibling-Based Admixture Studies

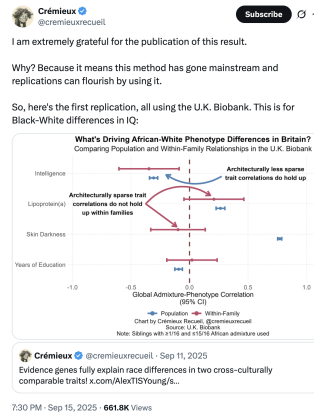
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1 Introduction

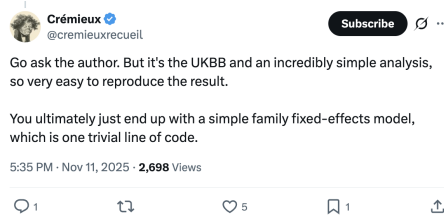
Sibling-based admixture studies leverage random variation in ancestry within sibling pairs to estimate the direct effect of ancestry on a given phenotype. While such studies are more capable of lending themselves to a causal interpretation than their population-wide counterparts, they require large samples of admixed sibling pairs to achieve adequate power to test hypotheses of interest. The goal of this paper is to demonstrate exactly why such high demands for statistical power arise in practice.

To motivate this goal, we will consider the [sibling-based admixture study](#) shared by the pseudonymous blogger Crémieux. This study purports to show that population-wide association between African ancestry and IQ is largely mediated by direct genetic effects.



2 Empirical Specification

Sibling-based admixture studies can be conducted using a sibship fixed effects specification. In this specification, the within-family effect of ancestry is estimated by the regression: $y_i = \beta_{WF}\pi_{i,j} + \mu_j + \epsilon_{i,j}$ where $\pi_{i,j}$ is the ancestry of individual i in family j and μ_j is a sibship fixed effect. Importantly, the analysis endorsed by Crémieux [uses](#) this specification.



Note that the sibship fixed effects specification can be re-written as a demeaned regression of the following form:

$$\begin{aligned} y_{i,j} - \bar{y}_j &= \beta_{WF} \cdot (\pi_{i,j} - \bar{\pi}_j) + (\epsilon_{i,j} - \bar{\epsilon}_j) \\ \widetilde{y}_{i,j} &= \beta_{WF} \cdot \widetilde{\pi}_{i,j} + \widetilde{\epsilon}_{i,j} \end{aligned} \quad (1)$$

Given a minimum effect size of interest of $\beta_{WF} = \beta$, the standard error of $\hat{\beta}_{WF}$ can be written in two ways:

$$\begin{aligned} SE(\widehat{\beta_{WF}}) &= SE(\widehat{\beta_{WF}}) \\ \frac{\sigma_{\tilde{\epsilon}}}{\sqrt{n}\sigma_{\tilde{\pi}}} &= \frac{\beta}{(Z_{1-\alpha/2} + Z_{1-\beta})} \end{aligned}$$

This equality can be used to derive the number of admixed sibling pairs needed to detect an effect of $\beta_{WF} = \beta$:

$$n = \frac{\sigma_{\tilde{\epsilon}}^2 \cdot (Z_{1-\alpha/2} + Z_{1-\beta})^2}{\beta^2 \cdot \sigma_{\tilde{\pi}}^2} \quad (2)$$

Consistent with standard practice, we assume $\alpha = 0.05$ (i.e. a false-positive rate of 5%) and $\beta = 0.2$ (i.e. a false-negative rate of 20%). This leaves 3 main unknowns for our sample size calculation:

- β : the minimum effect size of interest.
- $\sigma_{\tilde{\pi}}^2$: the within-family variance in major ancestry proportion.
- $\sigma_{\tilde{\epsilon}}^2$: the within-family residual variance in IQ.

For notation purposes, we assume ancestry exists on a 0 – 1 scale with 0 representing 0% ancestry in the focal ancestry group and 1 representing 100% ancestry in the focal ancestry group. We also assume that the dependent variable

is the population-standardized Fluid Intelligence Score in the UK Biobank and the minimum detectable effect represents the effect of a 100 percentage point increase in the focal ancestry proportion on the standardized Fluid Intelligence Score.

Importantly, these scaling choices do not affect the resulting power calculations. For example:

- If a 0 – 100 ancestry scale is used instead of a 0 – 1 scale, β^2 would be multiplied by $\frac{1}{100^2}$ and σ_{π}^2 would be multiplied by 100^2 , resulting in no net difference to the sample size.
- If an unstandardized Fluid Intelligence Score is used as the dependent variable in place of the population-standardized score, β^2 would be multiplied by $\frac{1}{\text{Var}(y)}$ and σ_{ϵ}^2 would be multiplied by $\frac{1}{\text{Var}(y)}$, resulting in no net difference to the sample size.

3 The Minimum Effect Size of Interest

The *minimum effect size of interest* is simply the smallest effect size that we are interested in powering our study to detect. The smaller the minimum is, the larger the required sample size becomes. Because there is no strict consensus in the hereditarian community on the exact contribution of genes to Black-White IQ disparities, we can conduct power calculations under three different assumptions: 50% contribution, 75% contribution, and 100% contribution. Under these conditions, the minimum effect sizes of interest should be $\beta \approx 0.68$ SD, 1.02 SD, and 1.35 SD, respectively. The rationale for these values is:

1. The analysis endorsed by Crémieux analyzes sibling pairs with African ancestry proportion between 6.25% and 93.75%. While Crémieux does not report the mean African ancestry proportion of this sample, Hu et al (2025) [provides](#) us with African ancestry proportions for roughly four-thousand admixed individuals in the UK Biobank.

Ancestry-aware PGS and correlation in effect sizes
 To understand the behavior of the PGS in African-ancestry individuals, we identified 8,003 UKB European-African admixed individuals whose sum of 27 UK or European ancestries and 4 sub-Saharan African ancestries is larger than 90% and sum of 4 sub-Saharan ancestries larger than 10%. We further binned these individuals into five ancestry bins based on their inferred African ancestry: [0.1, 0.35], 373 individuals, mean ancestry 0.207; [0.35, 0.55], 560 individuals, mean ancestry 0.46; [0.55, 0.78], 475 individuals, mean ancestry 0.707; [0.78, 0.95], 2,556 individuals, mean ancestry 0.89; and [0.95, 1], 4,039 individuals, mean ancestry 0.986. Individuals in the bin [0.95, 1] with near 100% African ancestry were only given an 'APGS' PGS in our analysis (see below), the results of which are shown in Fig. 4b,c and Extended Data Fig. 2 and Supplementary Figs. 2 and 12.

Using these ancestry proportions, we can conclude that the mean African ancestry proportion of admixed individuals in the analysis endorsed by Crémieux is likely around 74.3%.

2. I was unable to find a published estimate of the Black-White disparity in the Fluid Intelligence Score of the UK Biobank. That being said, hereditarian sources [place](#) the Black-White IQ gap in the United Kingdom at around 0.78 SD in magnitude. If we take this estimate at face value and allow for the possibility that it would become larger in a disproportionately educated sample (as [is the case](#) in the United States), then a plausible-to-generous reading would suggest that the Fluid Intelligence Score exhibits a 1 standard deviation Black-White disparity in the UK Biobank (because the UK Biobank is a disproportionately educated sample - see [van Alten et al \(2024\)](#) and [van den Berg et al \(2025\)](#)).
3. Assuming a Black-White IQ gap of 1 standard deviation, then, the direct effect of a 100 percentage point increase in African ancestry would need to be $1/0.743 = 1.35$ SD in magnitude for African ancestry to explain the entirety of the Black-White IQ gap in the United Kingdom.
4. If we are instead interested in detecting the ability of ancestry to explain 50% and 75% of the Black-White IQ gap in the United Kingdom, however, the minimum effect size of interest falls to $0.5 \cdot 1.35 = 0.68$ and $0.75 \cdot 1.35 = 1.02$, respectively.

Thus, $\beta \approx 0.68$ SD, 1.02 SD, and 1.35 SD under the 50%, 75%, and 100% conditions, respectively.

4 Within-Family Variation in Ancestry

Within-family variation in ancestry is the trickiest component to measure because:

1. There is currently no closed-form expression of this quantity in the population genetics literature as far as I'm aware.
2. Neither Crémieux nor any published study have reported the within-family variance of African ancestry among admixed sibships in the UK Biobank.
3. My attempts to derive a closed form expression of the within-family variance have been frustratingly unproductive.

Fortunately for our purposes, we do not need actually need a closed-form expression for the within-family variance to proceed. To demonstrate the shortcomings of the analysis endorsed by Crémieux, we need only place an *upper bound* on this quantity, and, as we shall see, an upper bound is much easier to produce.

I argue that, in a population with two-way admixture, within-family variance

in offspring ancestry is maximized under the following condition: *for every chromosome pair within each parent, the two homologous copies are ancestrally homogeneous and belong to entirely different ancestries (i.e., one homolog is 100% Ancestry A, and the other is 100% Ancestry B).*

The intuition behind this statement is two-fold:

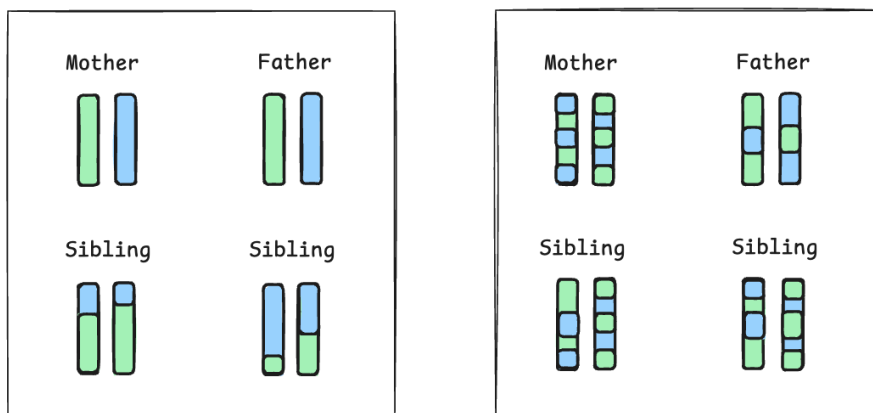
- **First, we can look to the variance of a Bernoulli distribution.**

When you flip a coin, the variance in the set of outcomes is maximized when the coin is fair. This is because, when the coin is unfair (i.e. biased towards heads or tails), outcomes cluster towards the favored side. Analogously, within-family variance in offspring ancestry is maximized when each parent has 50-50 ancestry. Ancestry proportions are not biased in one direction, so the spread of possible ancestry proportions is maximized.

- **Second, we can look to “The Law of Large Numbers” (LLN).**

The LLN states that “the average of the results obtained from a large number of independent random samples converges to the true value, if it exists.” When the parents’ chromosomes are homogeneous with respect to ancestry, we can think of each chromosome as one big ol’ ancestry block. When recombination chops up these chromosomes across generations, however, we have a large number of small ancestry blocks. The implication of the LLN is that, as the ancestry blocks become smaller and more numerous over time, the average ancestry proportion across all blocks will converge to the mid-parental ancestry proportion, decreasing the within-family variance in ancestry over time. In visual terms, within-family variance in ancestry depends on the number of ancestry blocks even when parental ancestry proportions are held fixed.

Both parental generations have 50-50 ancestry ...



... but the first household's offspring exhibit much greater variation!

We wish to derive a closed-form expression for the within-family variance of ancestry under these conditions.

Let i denote an individual in the population and a_i denote the ancestry proportion of i 's autosomes. Next, let $a_{i,m}$ denote the ancestry proportion of the autosomes that i inherited from his mother and $a_{i,f}$ denote the ancestry proportion of the autosomes that i inherited from his father. The ancestry proportion of i 's autosomes is equal to the average of these two values:

$$a_i = \frac{a_{i,m} + a_{i,f}}{2} \quad (3)$$

Because we are interested in deriving the within-family variance in ancestry, let i_1 and i_2 denote two siblings, $\bar{a}$ denote the average ancestry proportion of the two siblings, and $a_m = a_f = 0.5$ denote the ancestry proportions of i_1 and i_2 's parents under the variance-maximizing conditions. Then, we can derive an expression for the within-family variance of ancestry like so:

$$\begin{aligned} \text{Var}(a_{i_1} - \bar{a}) &= \text{Var}\left(a_{i_1} - \frac{a_{i_1} + a_{i_2}}{2}\right) \\ &= \text{Var}\left(\frac{a_{i_1} - a_{i_2}}{2}\right) \end{aligned}$$

Recall that, in this derivation, we are explicitly conditioning on the ancestral structure of each of the parent's chromosomes (i.e. that $a_m = a_f = 0.5$ and that each of the parent's chromosomes are ancestrally homogeneous). Conditional on this knowledge, the ancestry proportions of the siblings are uncorrelated because they are governed by independent meiotic processes. The variance of a summation thus becomes a summation of variances like so:

$$= \frac{1}{4}[\text{Var}(a_{i_1}) + \text{Var}(a_{i_2})]$$

The variance of each of sibling's ancestry should be equal to one another since the underlying mechanics of the meiotic processes remain the same between siblings. Thus, we can claim that $\text{Var}(a_{i_1}) = \text{Var}(a_{i_2}) = \text{Var}(a_i)$ and simplify like so:

$$\begin{aligned} &= \frac{1}{4}[\text{Var}(a_i) + \text{Var}(a_i)] \\ &= \frac{1}{2}\text{Var}(a_i) \end{aligned}$$

We can substitute Equation (3) to yield the following expression:

$$= \frac{1}{2}\text{Var}\left(\frac{a_{i,m} + a_{i,f}}{2}\right)$$

Because the inheritance of maternal vs paternal autosomes is governed by independent meiotic events, the variance of a summation becomes a sum of variances like so:

$$= \frac{1}{8} [Var(a_{i,m}) + Var(a_{i,f})] \quad (4)$$

To simplify the expression further, we will need to derive an expression for the variance of the ancestry proportion of autosomes inherited from a given parent. Without loss of generality, we will consider the variance corresponding to the maternally inherited autosomes $Var(a_{i,m})$.

Let $L_{m,c}$ denote the length of a given maternally inherited autosome c in Morgans. Each autosome can be said to be composed of many loci, each of which either belongs to the ancestry group or does not. We use the index j to denote a particular locus and the indicator variable ℓ_j to denote the ancestry at that locus.

$$\ell_j = \begin{cases} 1 & \text{if } j \text{ belongs to the ancestry group in question} \\ 0 & \text{if } j \text{ does not belong to the ancestry group in question} \end{cases}$$

The ancestry proportion of maternally inherited autosome c can thus be written like so:

$$a_{i,m,c} = \frac{1}{L_{m,c}} \int_0^{L_{m,c}} \ell_j dj$$

In other words, the ancestry proportion of each maternally inherited autosome is just the percentage of loci on that autosome which belong to the ancestry group.

By extension, the ancestry proportion of the maternally inherited autosomes is just a weighted average of the ancestry proportions for each autosome.

$$\begin{aligned} a_{i,m} &= \sum_{c=1}^{22} \frac{L_{m,c}}{L_m} \cdot \frac{1}{L_{m,c}} \cdot \int_0^{L_{m,c}} \ell_j dj \\ &= \sum_{c=1}^{22} \frac{1}{L_m} \cdot \int_0^{L_{m,c}} \ell_j dj \\ &= \frac{1}{L_m} \sum_{c=1}^{22} \int_0^{L_{m,c}} \ell_j dj \end{aligned}$$

The variance of the ancestry proportion of the maternally inherited autosomes can be written like so:

$$\begin{aligned}
Var(a_{i,m}) &= Var\left(\frac{1}{L_m} \sum_{c=1}^{22} \cdot \int_0^{L_{m,c}} \ell_j dj\right) \\
&= \frac{1}{L_m^2} Var\left(\sum_{c=1}^{22} \cdot \int_0^{L_{m,c}} \ell_j dj\right) \\
&= \frac{1}{L_m^2} \sum_{c_p=1}^{22} \sum_{c_q=1}^{22} Cov\left(\int_0^{L_{m,c_p}} \ell_j dj, \int_0^{L_{m,c_q}} \ell_j dj\right) \\
&= \frac{1}{L_m^2} \left[\sum_{c=1}^{22} Var\left(\int_0^{L_{m,c}} \ell_j dj\right) + \right. \\
&\quad \left. 2 \cdot \sum_{c_p=1}^{22} \sum_{c_q < c_p}^{22} Cov\left(\int_0^{L_{m,c_p}} \ell_j dj, \int_0^{L_{m,c_q}} \ell_j dj\right) \right] \\
&= \frac{1}{L_m^2} \left[\sum_{c=1}^{22} Var\left(\int_0^{L_{m,c}} \ell_j dj\right) + \right. \\
&\quad \left. 2 \cdot \sum_{c_p=1}^{22} \sum_{c_q < c_p}^{22} Cov\left(L_{m,c_p} \cdot m_{i,c_p}, L_{m,c_q} \cdot m_{i,c_q}\right) \right]
\end{aligned}$$

Because the length of a given autosome is a fixed constant, we can bring the length terms out of the covariance like so:

$$\begin{aligned}
&= \frac{1}{L_m^2} \left[\sum_{c=1}^{22} Var\left(\int_0^{L_{m,c}} \ell_j dj\right) + \right. \\
&\quad \left. 2 \cdot \sum_{c_p=1}^{22} \sum_{c_q < c_p}^{22} L_{m,c_p} L_{m,c_q} \cdot Cov\left(m_{i,c_p}, m_{i,c_q}\right) \right]
\end{aligned}$$

Conditional on knowing the maternal ancestry proportion $a_m = 0.5$ and the ancestral homogeneity of the maternal chromosomes, each inherited autosome's ancestry proportion is governed by independent meiotic events. Thus, the covariance of the ancestry proportions between two autosomes in the same individual is zero, and we can simplify like so:

$$= \frac{1}{L_m^2} \left[\sum_{c=1}^{22} Var\left(\int_0^{L_{m,c}} \ell_j dj\right) \right] \tag{5}$$

To simplify the equation further, we will need to derive the variance of the ancestry proportion of an arbitrary autosome c :

$$\begin{aligned} \text{Var}\left(\int_0^{L_{m,c}} \ell_j dj\right) &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} \text{Cov}(\ell_j, \ell_k) dj dk \\ &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} \mathbb{E}[\ell_j \ell_k] - \mathbb{E}[\ell_j] \mathbb{E}[\ell_k] dj dk \end{aligned}$$

Conditional on the ancestry proportion of the mother, the expected value of ancestry at a given loci on autosome c is simply the ancestry proportion of the mother.

$$\begin{aligned} &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} \mathbb{E}[\ell_j \ell_k] - a_m^2 dj dk \\ &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} P(\ell_j = 1 \cap \ell_k = 1) - a_m^2 dj dk \\ &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} P(\ell_j = 1 | \ell_k = 1) \cdot P(\ell_k = 1) - a_m^2 dj dk \\ &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} P(\ell_j = 1 | \ell_k = 1) \cdot a_m - a_m^2 dj dk \quad (6) \end{aligned}$$

To simplify the equation further, we will need to calculate the probability that a given locus j on autosome c belongs to the focal ancestry group given that another locus k on the same autosome does.

Note that the ancestry at locus j can only be the same as the ancestry at locus k if recombination has not split apart the two loci. If recombination has split apart the two loci, the loci belong to separate ancestry blocks, and the probability that j belongs the ancestry group in question is zero.

This claim necessarily raises the question: how can we calculate the probability that genetic recombination has *not* split apart two loci? Enter Haldane's mapping function.

Haldane's mapping function calculates the probability of genetic recombination occurring between two loci r given that the two loci are separated by a distance of d Morgans.

$$r = \frac{1}{2}(1 - e^{-2d})$$

This expression allows us to calculate the condition probability that j belongs to the ancestry group like so:

$$P(\ell_j = 1 | \ell_k = 1) = \frac{1}{2}(1 + e^{-2d})$$

We can now substitute the above result into Equation (6).

$$= \int_0^{L_{m,c}} \int_0^{L_{m,c}} \left(\frac{1}{2}(1 + e^{-2d}) \right) \cdot a_m - a_m^2 \, dj \, dk$$

Given $a_m = 0.5$, we can simplify like so:

$$\begin{aligned} &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} \frac{1}{4} e^{-2d} \, dj \, dk \\ &= \int_0^{L_{m,c}} \int_0^{L_{m,c}} \frac{1}{4} e^{-2|j-k|} \, dj \, dk \end{aligned}$$

We can make the integral easier to solve by simply solving for the case where $j > k$ and multiplying the resulting double integral by 2. Making this change requires us to swap the order of integration.

$$\begin{aligned} &= 2 \cdot \int_0^{L_{m,c}} \int_0^j \frac{1}{4} e^{-2(j-k)} \, dk \, dj \\ &= \frac{1}{2} \int_0^{L_{m,c}} \frac{1}{2} e^{-2j+2k} \Big|_0^j \, dj \\ &= \frac{1}{4} \int_0^{L_{m,c}} 1 - e^{-2j} \, dj \\ &= \frac{1}{4} \left(j + \frac{1}{2} e^{-2j} \Big|_0^{L_{m,c}} \right) \\ &= \frac{1}{4} \left[\left(L_{m,c} + \frac{1}{2} e^{-2L_{m,c}} \right) - \left(0 + \frac{1}{2} e^{-2 \cdot 0} \right) \right] \\ &= \frac{1}{4} \left(L_{m,c} + \frac{1}{2} e^{-2L_{m,c}} - \frac{1}{2} \right) \end{aligned}$$

We can now substitute the above result into Equation (5) to yield the variance of the maternally inherited autosomes.

$$\begin{aligned} Var(a_{i,m}) &= \frac{1}{L_m^2} \left[\sum_{c=1}^{22} Var \left(\int_0^{L_{m,c}} \ell_j \, dj \right) \right] \\ &= \frac{1}{L_m^2} \left[\sum_{c=1}^{22} \frac{1}{4} \left(L_{m,c} + \frac{1}{2} e^{-2L_{m,c}} - \frac{1}{2} \right) \right] \end{aligned}$$

From the above result, it is easy to see that the variance of the paternally inherited autosomes is equal to the following:

$$Var(a_{i,f}) = \frac{1}{L_f^2} \left[\sum_{c=1}^{22} \frac{1}{4} \left(L_{f,c} + \frac{1}{2} e^{-2L_{f,c}} - \frac{1}{2} \right) \right]$$

Lastly, we can substitute the above results into Equation (4) to yield the within-family variance of ancestry.

$$\begin{aligned} Var(a_i - \bar{a}) &= \frac{1}{8} [Var(a_{i,m}) + Var(a_{i,f})] \\ &= \frac{1}{8} \left[\frac{1}{L_m^2} \left[\sum_{c=1}^{22} \frac{1}{4} \left(L_{m,c} + \frac{1}{2} e^{-2L_{m,c}} - \frac{1}{2} \right) \right] + \right. \\ &\quad \left. \frac{1}{L_f^2} \left[\sum_{c=1}^{22} \frac{1}{4} \left(L_{f,c} + \frac{1}{2} e^{-2L_{f,c}} - \frac{1}{2} \right) \right] \right] \end{aligned}$$

The derivation is complete. Given the sex-specific lengths of the 22 autosomes in Morgans [reported](#) by Broman et al (1998) ...

- Chromosome 1: Female = 3.65 M, Male = 2.16 M
- Chromosome 2: Female = 3.31 M, Male = 2.09 M
- Chromosome 3: Female = 2.70 M, Male = 1.90 M
- Chromosome 4: Female = 2.64 M, Male = 1.60 M
- Chromosome 5: Female = 2.45 M, Male = 1.51 M
- Chromosome 6: Female = 2.54 M, Male = 1.31 M
- Chromosome 7: Female = 2.31 M, Male = 1.34 M
- Chromosome 8: Female = 2.24 M, Male = 1.13 M
- Chromosome 9: Female = 1.93 M, Male = 1.43 M
- Chromosome 10: Female = 2.11 M, Male = 1.38 M
- Chromosome 11: Female = 1.80 M, Male = 1.15 M
- Chromosome 12: Female = 2.14 M, Male = 1.28 M
- Chromosome 13: Female = 1.30 M, Male = 0.95 M
- Chromosome 14: Female = 1.55 M, Male = 1.17 M
- Chromosome 15: Female = 1.36 M, Male = 1.10 M
- Chromosome 16: Female = 1.69 M, Male = 1.01 M
- Chromosome 17: Female = 1.49 M, Male = 1.05 M
- Chromosome 18: Female = 1.56 M, Male = 0.97 M
- Chromosome 19: Female = 1.14 M, Male = 0.96 M
- Chromosome 20: Female = 1.21 M, Male = 0.81 M

- Chromosome 21: Female = 0.65 M, Male = 0.51 M
- Chromosome 22: Female = 0.74 M, Male = 0.49 M

... we get a within-family variance of ancestry approximately equal to 0.001291 in the variance-maximizing case.

Before proceeding to the concluding section, four points about the derivation are worth noting:

1. **Haldane’s mapping function is an approximation.** Haldane’s mapping function assumes crossover events occur independently of one another. In reality, however, crossover events exhibit *interference* (i.e. a crossover event at one locus makes the probability of a crossover event at a nearby locus less likely) and are more likely to occur in certain places in the genome (e.g. towards the ends of the chromosome in males and towards the center of the chromosome in females). For these reasons, Haldane’s mapping function [appears](#) to under-estimate recombination rates in human beings ([Kivikoski et al 2023](#)).
2. **The derivation applies to autosomes only.** Throughout this derivation, we have focused solely on the 22 autosomes of the human genome. The primary reason for doing so is two-fold. First, ancestry exhibits sex-specific inheritance patterns. Among African-Americans, for example, women carry higher proportions of African ancestry on average than males. Second, for the purposes of this paper, we are interested in the possible contribution of ancestry to differences in cognitive ability between populations, an arena where sex chromosomes are unlikely to matter.
3. **The derivation is confirmed by simulations.** When [simulating](#) sibling pairs under the variance-maximizing conditions and the assumptions of Haldane’s mapping function, the within-family variance in ancestry is approximately 0.001291, exactly what is predicted by the analytical derivation.

5 Within-Family Residual Variation in IQ

Let i_1 and i_2 denote two siblings in sibship j . Given a sibling correlation in IQ of ρ , the within-family variation in IQ can be written like so:

$$\begin{aligned}
\sigma_y^2 &= Var(y_{i_1,j} - \bar{y}_j) \\
&= Var\left(y_{i_1,j} - \frac{y_{i_1,j} + y_{i_2,j}}{2}\right) \\
&= Var\left(\frac{y_{i_1,j} - y_{i_2,j}}{2}\right) \\
&= \frac{1}{4}[Var(y_{i_1,j}) + Var(y_{i_2,j}) - 2Cov(y_{i_1,j}, y_{i_2,j})]
\end{aligned}$$

We assume $Var(y_{i_1,j}) = Var(y_{i_2,j}) = \sigma_y^2$, the variance of IQ in the general population.

$$\begin{aligned}
&= \frac{1}{4}[\sigma_y^2 + \sigma_y^2 - 2Cov(y_{i_1,j}, y_{i_2,j})] \\
&= \frac{1}{4}[2\sigma_y^2 - 2\rho\sigma_y^2] \\
&= \frac{\sigma_y^2(1 - \rho)}{2}
\end{aligned}$$

Because we assume the dependent variable has already been population-standardized, the variance term is equal to 1 and we have:

$$= \frac{1 - \rho}{2}$$

Table E.2 of Markel et al (2025) [report](#) the sibling correlation in the UK Biobank's Fluid Intelligence Score to be $\rho = 0.287$.

Table E.2 (continued): Results for each dataset

Phenotype	UKB				UKB				UKB			
	N_{fam}	N^*	r^2_{F}	P_{FAM}	N_{fam}	N^*	r^2_{F}	P_{FAM}	N_{fam}	N^*	r^2_{F}	P_{FAM}
<i>Child & family and educational</i>												
Cognitive performance	127	6,601	0.288	6.0e-107	1,388	12,112	0.133	1.1e-05	1,689	12,601	0.141	8.1e-06
FA	9,473	49,862	0.286	6.0e-107	10,949	53,111	0.133	1.1e-05	12,701	63,900	0.140	8.1e-06
Child & labor market												
Employment	8,169	41,841	0.287	6.0e-107	9,361	47,811	0.133	1.1e-05	10,949	53,111	0.141	8.1e-06
Log health income	1,171	6,024	0.288	6.0e-107	1,388	12,112	0.133	1.1e-05	1,689	12,601	0.141	8.1e-06
Occupational Status	1,171	6,024	0.288	6.0e-107	1,388	12,112	0.133	1.1e-05	1,689	12,601	0.141	8.1e-06

Combining this result with the above claim, the within-family variance of IQ is:

$$\begin{aligned}
\sigma_y^2 &= \frac{1 - \rho}{2} \\
&= \frac{1 - 0.287}{2} \\
&= 0.3565
\end{aligned}$$

The within-family *residual* variation in IQ is equal to the following:

$$\sigma_{\epsilon}^2 = \sigma_y^2 - \beta_{WF}^2 \sigma_{\pi}^2$$

Substituting in the results of our previous derivations yields the following residual variance in the 50% case ...

$$\begin{aligned}
&= 0.3565 - (0.68)^2 \cdot 0.001291 \\
&\approx 0.3559
\end{aligned}$$

... the following residual variance in the 75% case ...

$$\begin{aligned}
&= 0.3565 - (1.02)^2 \cdot 0.001291 \\
&\approx 0.3552
\end{aligned}$$

... and the following residual variance in the 100% case ...

$$\begin{aligned} &= 0.3565 - (1.35)^2 \cdot 0.001291 \\ &\approx 0.3541 \end{aligned}$$

6 Conclusion

We now have all the parameter values necessary to carry out the sample size calculation.

First, we have the sample size needed to detect an effect which explains ~50% of the gap.

$$\begin{aligned} n &= \frac{\sigma_\epsilon^2 \cdot (Z_{1-\alpha/2} + Z_{1-\beta})^2}{\beta^2 \cdot \sigma_\pi^2} \\ &= \frac{0.3559 \cdot (1.96 + 0.84)^2}{(0.68)^2 \cdot 0.001291} \\ &\approx 4670 \end{aligned}$$

Second, we have the sample size needed to detect an effect which explains ~75% of the gap.

$$\begin{aligned} n &= \frac{\sigma_\epsilon^2 \cdot (Z_{1-\alpha/2} + Z_{1-\beta})^2}{\beta^2 \cdot \sigma_\pi^2} \\ &= \frac{0.3552 \cdot (1.96 + 0.84)^2}{(1.02)^2 \cdot 0.001291} \\ &\approx 2070 \end{aligned}$$

Third, we have the sample size needed to detect an effect which explains ~100% of the gap.

$$\begin{aligned} n &= \frac{\sigma_\epsilon^2 \cdot (Z_{1-\alpha/2} + Z_{1-\beta})^2}{\beta^2 \cdot \sigma_\pi^2} \\ &= \frac{0.3541 \cdot (1.96 + 0.84)^2}{(1.35)^2 \cdot 0.001291} \\ &\approx 1180 \end{aligned}$$

In other words, in the variance-maximizing case, at least one-thousand sibling pairs are needed to test the hereditarian hypothesis with adequate statistical power. This result is also confirmed by [power simulations](#). How many sibling pairs did the analysis endorsed by Crémieux include, then? According to the original author of the analysis, just 175 sibling pairs!



Thus, the analysis cannot produce reliable inferences. Even in the variance-maximizing case, the UK Biobank does not contain enough admixed sibling pairs to test the hereditarian hypothesis with adequate power.

Furthermore, the true sample size requirement is almost certainly likely to be drastically larger than one-thousand in practice. The variance-maximizing conditions assume parental ancestry proportions are exactly 50-50 and only one generation of recombination has split chromosomes into distinct ancestry blocks.

In practice, however, admixed individuals with African ancestry in the UK Biobank have a mean African ancestry proportion of around 75% (as shown in Section 4) and have undergone several generations of recombination, owing to the fact that they are descended from West Africans forcibly transferred to the UK as part of the Transatlantic Slave Trade (Honorato-Mauer et al 2025).

Our findings trace the African ancestral origins of most two-way EUR-AFR admixed UKB individuals to West and Central-West Africa, with a smaller portion tracing back to East Africa. These patterns are consistent with historical records of the transatlantic slave trade from the 16th to 19th centuries. We are now fitting demographic models to the local ancestry tract distributions of admixed African-descent UKB individuals to better characterize their demographic history for use in downstream efforts.

7 Appendix: Comparison with Wang et al (2026)

Wang et al (2026) conduct a within-family admixture study using the population of admixed individuals in the Mexico City Prospective Study. Supplementary Note 2 provides the following power calculations.

Supplementary Note 2: Statistical power to detect a within-family ancestry effect

Statistical power to detect a within-family ancestry effect depends on the effect size and its sampling variance, the latter being a function of the sample size and the within-family variance in ancestry proportions. We consider the simplest case of two ancestries and N independent families each with 2 siblings. Let ρ be the sibling phenotypic correlation and consider the regression $y_{i1} - y_{i2} = b_w(\pi_{i1} - \pi_{i2}) + \epsilon_{i1} - \epsilon_{i2}$, with $\text{var}(\epsilon_{i1} - \epsilon_{i2}) \approx 2(1-\rho) = \text{var}(y_{i1} - y_{i2})$. The expectation of $\hat{b}_w$, the ordinary least squares estimator of b_w , is $E(\hat{b}_w) = \beta$ and its sampling variance is, approximately, $\text{var}(\hat{b}_w) = 2(1-\rho) / (N2\sigma_d^2) = (1-\rho) / (N\sigma_d^2)$. The non-centrality-parameter (NCP) of a χ^2 test of $\beta = 0$ is $\text{NCP} = \beta^2 / \text{var}(\hat{b}_w) = N\sigma_d^2\beta^2 / (1-\rho)$. To calculate statistical power, an experimenter has to define the false positive rate (α_0), the value of β , the number of sibling pairs N , sibling correlation ρ and the within-family variance in ancestry proportion σ_d^2 . If we define $F(x, \kappa)$ as the cumulative density function of a χ^2 distribution with 1 degree of freedom with NCP equal to κ , and $Q(p)$ as the quantile function of a central χ^2 distribution with 1 degree of freedom and probability p , then the power of detection (Power) is

$$\text{Power} = 1 - F(Q(1-\alpha_0), N\sigma_d^2\beta^2/(1-\rho))$$

Of note, the above calculations are based on σ_d^2 , the variance of the deviation from the *parental* mean ancestry proportion. Because the power analysis conducted in this document defines the within-family variance of ancestry as the variance of the deviation from the *sibship* mean ancestry proportion, we will need to show how the former is mathematically related to the latter if we wish to compare the two power calculations side-by-side.

Let $a_{i_1}, a_{i_2}, \dots, a_{i_n}$ represent the ancestry proportions of n siblings in the same family. We can write each sibling's ancestry proportion as a sum of the parental ancestry mean a_p and the sibling-specific deviation d_i :

$$\begin{aligned} a_{i_1} &= a_p + d_{i_1} \\ a_{i_2} &= a_p + d_{i_2} \\ &\dots \\ a_{i_n} &= a_p + d_{i_n} \end{aligned}$$

We can then re-write the variance of the deviation from the *sibship* mean ancestry proportion like so:

$$\begin{aligned}
\text{Var}\left(a_{i_1} - \frac{1}{n} \sum_j a_{i_j}\right) &= \text{Var}\left((a_p + d_{i_1}) - \frac{1}{n} \sum_j (a_p + d_{i_j})\right) \\
&= \text{Var}\left((a_p + d_{i_1}) - (a_p + \frac{1}{n} \sum_j d_{i_j})\right) \\
&= \text{Var}\left(d_{i_1} - \frac{1}{n} \sum_j d_{i_j}\right) \\
&= \text{Var}\left(\frac{n-1}{n} \cdot d_{i_1} - \frac{1}{n} \sum_{j \neq 1} d_{i_j}\right)
\end{aligned}$$

Because each sibling's deviation from the parental mean ancestry proportion is governed by independent meiotic processes, the covariance of the deviations is zero, and we can simplify like so:

$$\begin{aligned}
&= \frac{(n-1)^2}{n^2} \text{Var}(d_{i_1}) + \frac{1}{n^2} \sum_{j \neq i} \text{Var}(d_{i_j}) \\
&= \frac{(n-1)^2}{n^2} \sigma_d^2 + \frac{n-1}{n^2} \sigma_d^2 \\
&= \frac{n-1}{n} \sigma_d^2
\end{aligned}$$

Thus, when we have only $n = 2$ siblings per family (as assumed by the power calculations in this document), the variance of deviation from the *sibship* mean ancestry proportion is half of the variance of the deviation from the *parental* mean ancestry proportion.

What happens, then, when we conduct an apples-to-apples comparison between the power calculations in this document and those in Wang et al (2026)? Well if we ...

- Use a within-family variance of $2 \cdot 0.001291 = 0.002582$ to reflect the maximum possible within-family variance of ancestry as defined in Wang et al (2026) (i.e. a standard deviation of 5.1 percentage points)
- Re-use the same sibling correlation of 0.287 reported in Markel et al (2025).
- Use $\beta = 1.35$ SD to mimic the 100% contribution condition.
- Use $n \approx 1180$ to reflect the sample size requirement yielded by the power calculations in this document.

... we can see that the power calculations are corroborated! The [R code](#) provided by the Wang et al (2026) authors estimates statistical power to be 80%

under these conditions, exactly what our calculations aimed for.

```
+ #####
+ ## Example
+ #####
+ power_sibdiff(
+   n   = 1180,
+   r   = 0.287,
+   VAR_w = 0.001291*2,
+   beta = 1.35,
+   alpha = 0.05
+ )
[1] 0.7969305
```

Separately, the Wang et al power formula provides us with the ability to understand under what conditions an $n = 175$ within-family admixture study could achieve adequate power to test the hereditarian hypothesis.

$$\text{Power} = 1 - F\left(Q(1 - \alpha_0), \frac{N\sigma_d^2\beta^2}{1 - \rho}\right)$$

Given a significance threshold of $\alpha_0 = 0.05$ and a desired power of 80%, we can substitute like so:

$$\begin{aligned} 0.8 &= 1 - F\left(3.841, \frac{N\sigma_d^2\beta^2}{1 - \rho}\right) \\ 0.2 &= F\left(3.841, \frac{N\sigma_d^2\beta^2}{1 - \rho}\right) \end{aligned}$$

The value of the non-centrality parameter that satisfies this equality is 7.787826. This claim can be verified using the R code `pchisq(3.841, 1, 7.787826)`. Thus, we have:

$$7.787826 = \frac{N\sigma_d^2\beta^2}{1 - \rho}$$

Given a sample size of 175 sibling pairs, a maximum within-family variance of 0.002582, and a 1.35 SD effect for a 100 percentage point increase in African ancestry, we can solve for the implied sibling correlation like so:

$$\begin{aligned}
7.787826 &= \frac{175 \cdot 0.002582 \cdot 1.35^2}{1 - \rho} \\
1 - \rho &= \frac{175 \cdot 0.002582 \cdot 1.35^2}{7.787826} \\
\rho &= -\frac{175 \cdot 0.002582 \cdot 1.35^2}{7.787826} + 1 \\
\rho &= 0.894
\end{aligned}$$

In other words, for $n = 175$ sibling pairs to have adequate power to detect a 1.35 SD effect of a 100 percentage point increase in ancestry, the sibling correlation in IQ must be at least $r \approx 0.9$. Because such a condition is completely unrealistic, it follows that the study shared by Crémieux is highly unlikely to have adequate power.

Lastly, Wang et al provide a nice illustration of how small within-family variation in ancestry is in real-world admixed populations. In their sample, the within-family variance does not exceed 0.000422 (i.e. a standard deviation of 2.1 percentage points), well below the 0.002582 derived in this document (i.e. a standard deviation of 5.1 percentage points).

Variance	Parental and offspring genotypes ^a			Offspring genotypes only ^a		
	Between	Within	λ	Between	Within	λ
	family	family		family	family	
IAM	2.61E-02	4.22E-04	0.016	2.79E-02	3.85E-04	0.014
EUR	2.13E-02	3.25E-04	0.015	2.28E-02	3.11E-04	0.014
AFR	4.75E-04	3.56E-05	0.075	5.41E-04	3.45E-05	0.064
EAS	2.33E-04	4.33E-05	0.186	1.35E-04	3.06E-05	0.227